National Institute of Technology Kurukshetra

Department of Computer Engineering

SOFTWARE PROJECT MANAGEMENT (CSPE 318/ITPE 318)

Date of Exam – 22-Feb-2025

Time: 03:00 PM-03:50 PM [50 min] **Max Marks:15**

Attempt all questions

Note: Mention your question number properly and Don't Write Irrelevant Stories in your ANSWERS.

1. Fill in the Blanks
   a. Full Form of SEL Model is _____ and WF Model is _____ **(01)**
   b. Defect Rate is measured using the formula _____. **(01)**
   c. LOC-COUNT estimation is indirect estimation? True or False _____. **(01)**
   d. Unit of Software Size can be _____, _____, _____ and _____. **(01)**
   e. A software project is estimated to have 40,000 LOC. The team will consist of 4 programmers, who can develop 80 LOC per day. The project will run for 20 weeks. What is the total estimated effort for the project in person-months? _____ **(01)**
2. What are the different ways of categorizing the software projects and stakeholders? What are the steps involved in step-wise project planning, explain with a diagram. **(4)**
3. (a) Explain the COCOMO Model. Given the table below, compute effort, duration and team size for organic, semidetached, and embedded software projects. The software is of 20KLOC. **(4)**

Software Project	a_b	b_b	c_b	d_b
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

(b) The development effort for a software project is 300 persons-months. The empirically determined constant (K) is 0.3. The complexity of the code is equal to 8 (quite high value). Calculate the total effort (M) if,

The maintenance team has good level of understanding of the project (d = 0.9).
The maintenance team has poor understanding of project (d = 0.1) **(2)**